

Git

19 June 2026 18:26

What is Git?

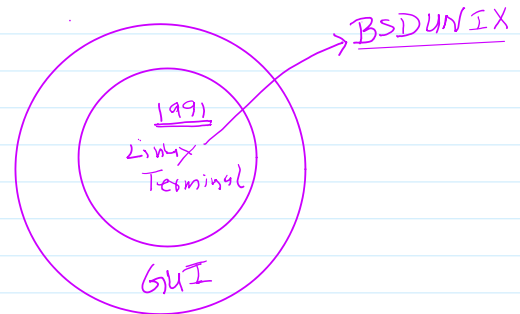
Git is a **distributed version control system (DVCS)** used to track changes in source code and manage software development projects.

It was created by Linus Torvalds in 2005 for the development of the Linux kernel.

In simple terms what **git** is?

Git is a tool that helps developers:

- Track changes in files
- Maintain different versions of a project
- Collaborate with other developers
- Restore previous versions when needed



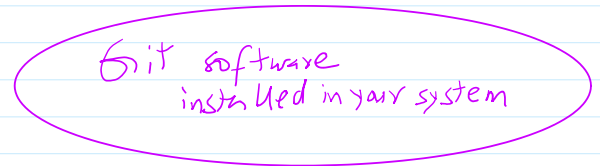
Why do we need Git?

Imagine you are developing a project.

Without Git:

Project

```
|  
|--->Project_final.py  
|--->Project_final_new.py  
|--->Project_final_latest.py  
|--->Project_final_Really_Final.py
```



Managing versions becomes difficult.

With Git

Project

```
|--->git repository  
|   |--->Version 1  
|   |--->Version 2  
|   |--->Version 3  
|   |--->Version 4
```

Version 1 → A
Version 2 → A ⊕ B

Git automatically tracks every change.

Features of Git

- Version Control:** Tracks every modification made to files.
- Distributed System:** Every developer has a complete copy of the repository.
- Fast Performance:** Operations are performed locally.
- Branching:** To allow development of new features without affecting the main project.
- Backup and Recovery:** Previous version can be restored easily.
- Collaboration:** Multiple developers can work on same project simultaneously.

Git Architecture

Working Directory

```
|  
V  
git add  
|  
V
```



Git Terminologies

Repository(Repo): A repository is a storage location of your project.

Project Files + Git History = Repository

☐ Name	Date modified	Type	Size
.git	19-06-2026 19:14	File folder	
Code	19-06-2026 19:13	File folder	
Documents	19-06-2026 19:13	File folder	

Repository

Commit: A commit is a snapshot of your project at a specific point in time.

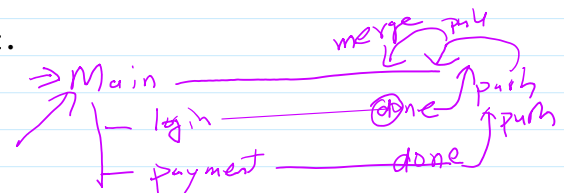
Example:

Commits on May 28, 2026
Discussed prime number and coprime concepts sigmapl-academy committed 3 weeks ago a5be979
Commits on May 22, 2026
Discussed nested looping concept sigmapl-academy committed last month 77979fa
Commits on May 13, 2026
Repetition for loop discussed with homework sigmapl-academy committed on May 13 d09162f

Branch: A branch is an independent line of development.

```

Main
|-->feature-login
|-->feature-payment
  
```



Merge: Combines changes from different branches.

```

Feature-login
|
V
Merge
|
V
main
  
```

git init

```
MINGW64:/c/Users/bluej/Documents/Students/ShivRay/NewIdea
bluej@LAPTOP-106N11QP MINGW64 ~/Documents/Students/ShivRay/NewIdea (master)
$ git init
Initialized empty Git repository in C:/Users/bluej/Documents/Students/ShivRay/NewIdea/.git/
```

↑
hidden folder.